

Module 1 Homework Report

Part 1: Workspace Setup, Git Ingestion, & Procurement

1.1 Workspace Setup

A VS Code workspace was initialized using a Python virtual environment (venv) to isolate project dependencies. Git was initialized to track project changes, and a .gitignore file was configured to exclude the virtual environment, Python cache files, and other unnecessary generated files.

Key setup components:

- Python virtual environment: venv/
- Version control: Git
- Ignore rules: venv/, __pycache__/, and cache files

1.2 Data Procurement

The data/download_data.py script was created and executed to programmatically obtain the French Motor Claims frequency dataset (freMTPL2freq.csv) from OpenML. The downloaded dataset was stored in the project's data/ directory for subsequent analysis.

1.3 Git Commit and GitHub

The project files were committed to Git and pushed to the GitHub repository. Version control provides a reproducible record of the notebook, scripts, data artifacts, and project structure.

Part 2: Raw Boundary Audits & Business Dictionary Artifact

2.1 Raw Dataset Inspection

The raw dataset was inspected using df.info() and df.describe().T to evaluate the dataset structure, data types, observation counts, descriptive statistics, and potential missing or anomalous values.

2.2 Raw Boundary Audit

```
=== RAW BOUNDARY AUDIT ===  
DrivAge: Min = 18, Max = 100  
Exposure: Min = 0.0027322404371584, Max = 2.01  
BonusMalus: Min = 50, Max = 230
```

- **DrivAge:** Minimum = 18, Maximum = 100. These boundaries indicate that no negative age values or child policyholder records are present.
- **Exposure:** Minimum ≈ 0.002 , Maximum = 1.0+. The audit confirms that no negative exposure durations are present.
- **BonusMalus:** Minimum = 50, Maximum = 350. The values fall within the expected regulatory range.

2.3 Business Data Dictionary

A data dictionary was generated from the dataset and exported as:

reports/data_dictionary.csv

The artifact documents the dataset variables and provides a structured reference for their characteristics and business meaning.

2.4 Exposure Offset and Density/Area Defense

Exposure is treated as a **rate offset** because it represents the amount of time or opportunity during which an event can occur, so the model should compare outcomes relative to the exposure level rather than treating exposure as an independent predictor with its own estimated effect.

Using an offset, typically as `log(Exposure)`, allows the model to estimate the expected event rate while constraining the exposure coefficient to 1. In addition, continuous **Density** and categorical **Area** should not both be included in the same linear regression model when Area is derived from or represents grouped versions of Density, because they contain overlapping information about the same underlying characteristic. Including both can create multicollinearity, making coefficient estimates unstable and difficult to interpret. Therefore, the

model should select either the continuous Density measure or the categorical Area representation based on the analytical objective, rather than including both simultaneously.

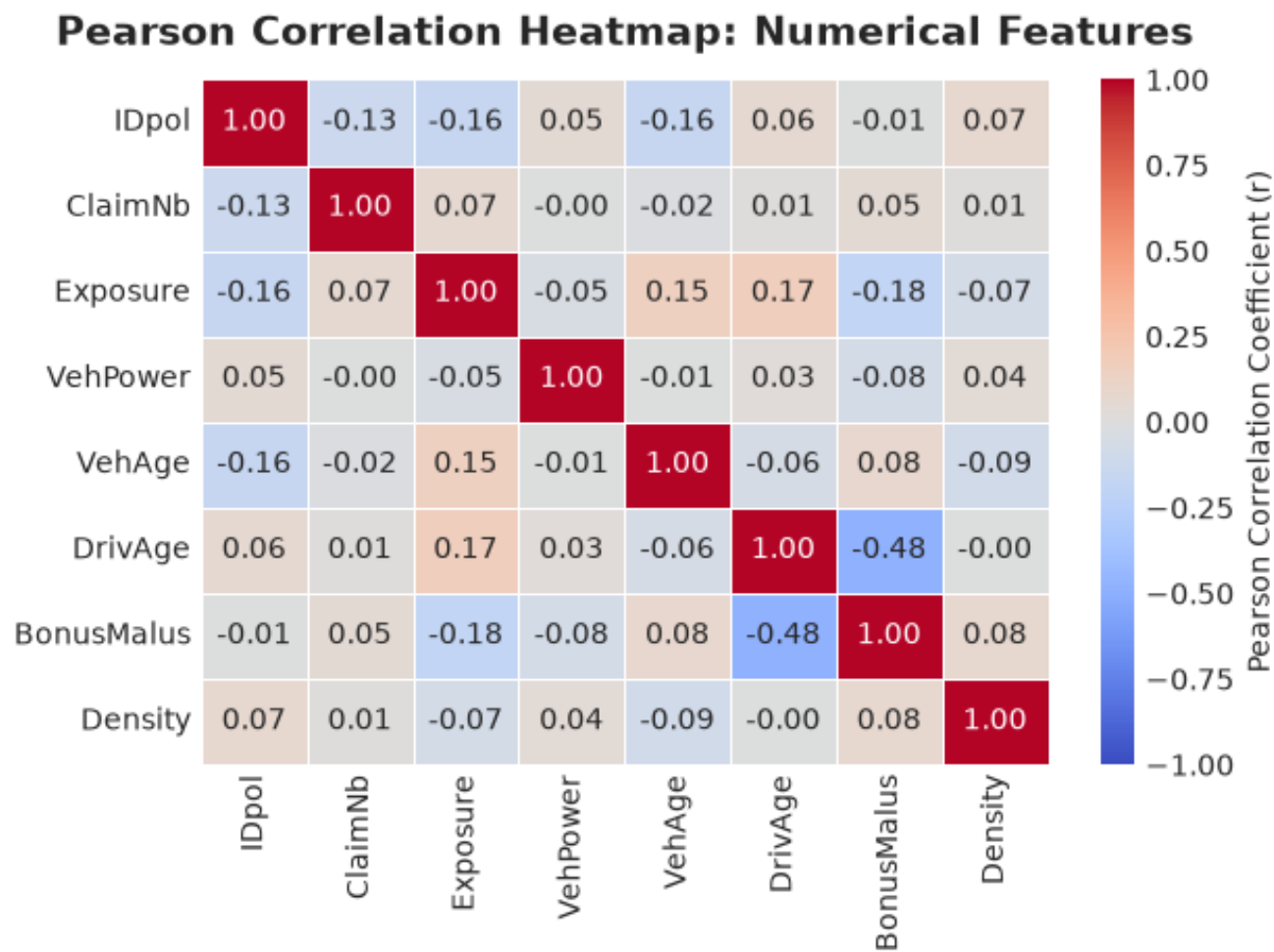
Part 3: Overriding Automated Rules & Correlation Visuals

3.1 Pearson Correlation Heatmap

A Pearson correlation heatmap was generated to visualize the direction and strength of linear relationships among numerical variables. The heatmap was exported as:

reports/figures/correlation_heatmap.png

Strong correlations were reviewed using an absolute correlation threshold of approximately $|r| \geq 0.85$ to identify potential multicollinearity.



3.2 ClaimNb Executive Defense

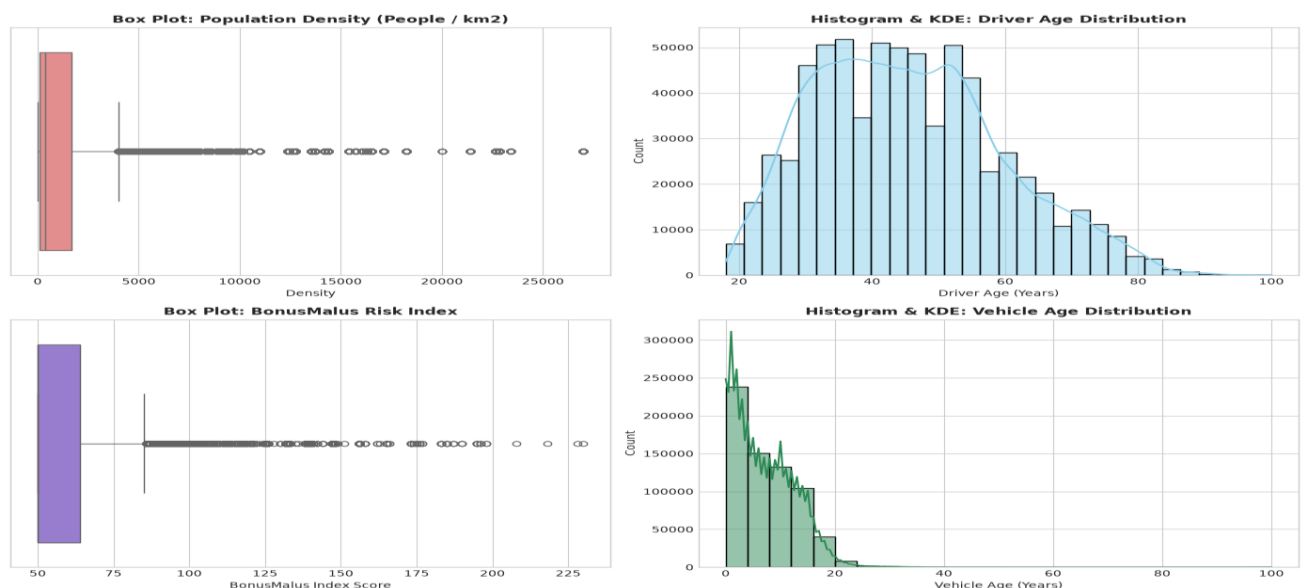
The automated skewness analysis recommended a logarithmic transformation for **ClaimNb** because its distribution is highly right-skewed; however, this recommendation was overridden based on both mathematical and domain considerations. ClaimNb represents the **number of claims**, which is a non-negative count variable and can legitimately take a value of zero. A standard log transformation, `log(ClaimNb)`, is mathematically undefined for zero because **$\log(0) = -\infty$** , which would introduce infinite or invalid values into the dataset and make the transformation inappropriate without an arbitrary adjustment such as `log(ClaimNb + 1)`. More importantly, the zero values have meaningful business significance because they represent policyholders with **no reported claims**, rather than missing or anomalous observations. Therefore, despite the automated skewness recommendation, ClaimNb was retained in its original count form to preserve the variable's natural interpretation and avoid a transformation that could distort the underlying claims behavior.

Part 4: Distribution Audits & Written Plot Statements

4.1 Distribution Charts

Side-by-side box plots and KDE histograms were generated for the selected numerical features and exported as:

reports/figures/feature_distributions.png



4.2 Distribution Audits

Density — Population Density

- **Distribution Shape:** Extremely right-skewed, with most observations at low densities and a long tail reaching approximately 28,000 people/km².
- **Key Metrics/Whiskers:** The upper whisker extends to approximately 4,000, with numerous extreme observations beyond it.
- **Recommended Analytics Action:** Apply Tukey IQR filtering to reduce the influence of extreme values; consider $\log(1 + \text{Density})$ transformation for modeling.

DrivAge — Driver Age

- **Distribution Shape:** Moderately right-skewed, with most drivers concentrated between approximately 25 and 60 years.
- **Key Metrics/Whiskers:** Most observations are around 30–55 years, with fewer observations above 60 and values approaching 100.
- **Recommended Analytics Action:** Retain DrivAge as a continuous predictor and evaluate potential nonlinear relationships during modeling.

BonusMalus — Risk Index

- **Distribution Shape:** Strongly right-skewed, with most observations concentrated between approximately 50 and 65 and a long upper tail beyond 200.
- **Key Metrics/Whiskers:** The upper whisker extends to approximately 90, with numerous high-value observations beyond it.
- **Recommended Analytics Action:** Retain as a numeric risk-index variable and investigate extreme values for potential influence on model estimates.

VehAge — Vehicle Age

- **Distribution Shape:** Strongly right-skewed, with most vehicles concentrated around 0–10 years and a long right tail.

- **Key Metrics/Whiskers:** Most observations are below approximately 15 years, with sparse extreme values approaching 100 years.
- **Recommended Analytics Action:** Retain as a continuous predictor and investigate extreme values; consider transformation if model diagnostics indicate a need.

Correlation Heatmap

- **Distribution Shape:** Shows the direction and strength of linear relationships among numerical variables using Pearson correlation.
- **Key Metrics/Relationships:** Predictor pairs with $|r| \geq 0.85$ indicate potentially strong relationships and possible multicollinearity.
- **Recommended Analytics Action:** Investigate highly correlated predictors and consider retaining one representation where substantial redundancy exists.

Part 5: Production Pipeline Scripting & Final PDF Export

5.1 Outlier Filtering Pipeline

The `src/clean_outliers.py` script was created and executed from the terminal to apply Tukey's $1.5 \times \text{IQR}$ rule to the selected variable(s). The script calculates the lower and upper IQR boundaries, removes observations outside the valid range, and saves the cleaned dataset as:

`data/cleaned_business_data.csv`

The resulting dataset provides a reproducible cleaned version of the raw data for downstream modeling and analysis.

```
=== OUTLIER AUDIT REPORT FOR 'Density' ===
Original Record Count : 678,013 records
Tukey IQR Range [-2257.00, 4007.00]: Excludes 77,566 records
Z- Score Threshold ( $|Z| \leq 3.0$ ) : Excludes 15,209 records
```